Electric Current

Current Electricity · Class 12

1 Definition

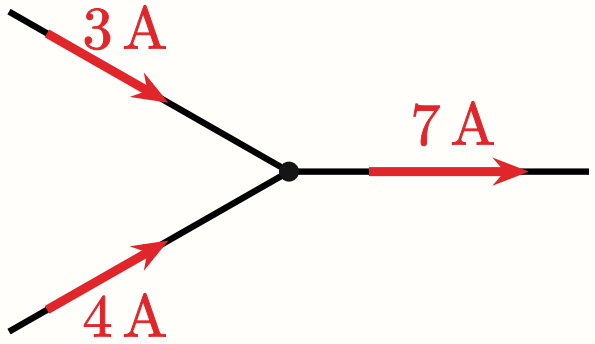
$$I = \frac{dq}{dt}, \qquad I_{avg} = \frac{\Delta q}{\Delta t}$$

q: net charge through a cross-section · $1\,\mathrm{A} = 1\,\mathrm{C\,s^{-1}}$ · steady: $q = It$

Example

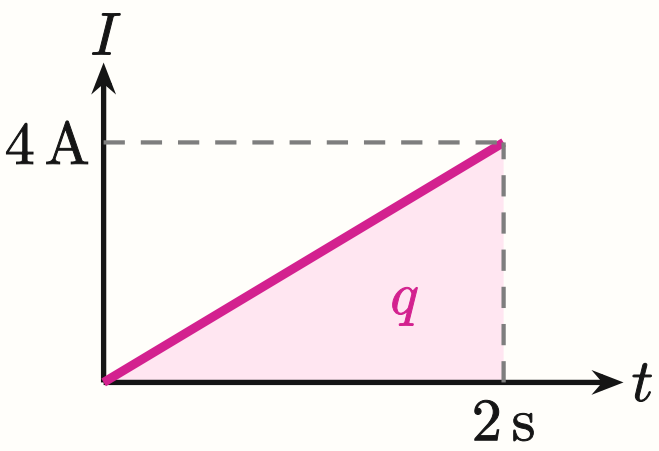
Steady $5\,\mathrm{A}$ for $2\,\mathrm{min}$: $q = 600\,\mathrm{C}$, i.e. 3.75×10^{21} electrons.

2 Current is a scalar



- Direction: that of POSITIVE charge flow (electrons drift the other way).
- At a junction currents add algebraically, at any angle: $7\,\mathrm{A}$, not $5\,\mathrm{A}$.

3 Charge from the I–t graph



- $q = \int I\,dt$ = area under the I–t graph.
- Here: $q = \frac{1}{2}(4)(2) = 4\,\mathrm{C}$.

Golden points

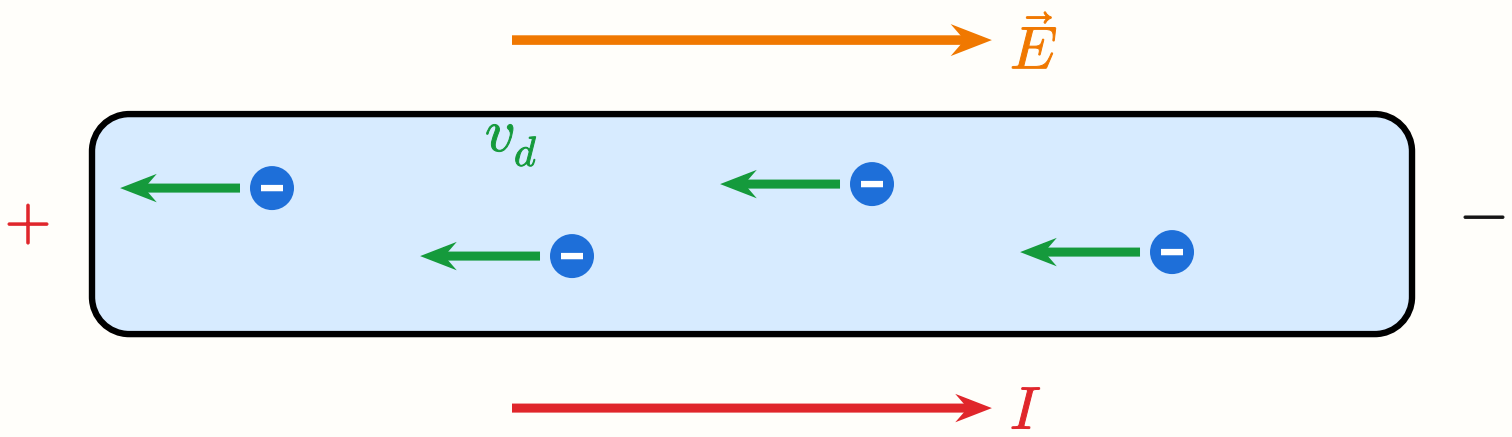
★ Charge q circling f times per second: $I = qf = q\omega/2\pi$. H-atom electron (ground state), $f = 6.6 \times 10^{15}\,\mathrm{Hz}$: $I \approx 1.06\,\mathrm{mA}$.

Free Electrons and Drift

1 No field: random motion

- Free electrons move randomly at thermal speeds $\sim 10^5\,\mathrm{m\,s^{-1}}$, colliding with the vibrating ions.
- No potential difference across the wire: AVERAGE velocity zero, no net current.

2 Field along the wire: drift



$$\vec{v}_d = -\frac{e\vec{E}}{m}\tau$$

τ: relaxation time, mean time between collisions ($\sim 10^{-14}\,\mathrm{s}$) · e, m: electron charge, mass · electrons drift OPPOSITE to $\vec{E}$

Golden points

★ The bulb glows at once: the field is set up along the whole wire almost at the speed of light, and all free electrons start drifting together.

★ Carriers: free electrons (metals), ions (electrolytes), electrons and ions (ionised gases), electrons and holes (semiconductors).

Drift Speed and Current

1 Current from drift

$$I = n\,e\,A\,v_d$$

n: free electrons per m^3 · A: cross-section · in time Δt the charge $(nAv_d\Delta t)\,e$ crosses A

Example

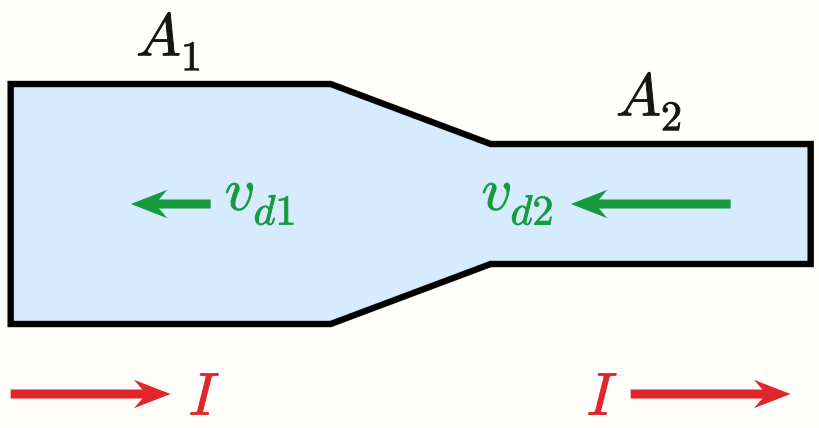
Copper, $n = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$, $A = 1.0\,\mathrm{mm}^2$, $I = 1.5\,\mathrm{A}$:
$v_d \approx 1.1 \times 10^{-4}\,\mathrm{m\,s}^{-1}$ *(about* $2.5\,\mathrm{h}$ *to drift* $1\,\mathrm{m}$*).*

2 Fixed voltage across a wire

$$v_d = \frac{e\tau}{m}\frac{V}{l}$$

$E = V/l$ uniform · same material and temperature · independent of A: double V, double v_d; double l, half v_d

3 Wire of changing cross-section



- *Steady current: the SAME I through every section.*
- *Narrow part: larger J, E, v_d.* $A_1 = 2A_2$: $J_2 = 2J_1$, $v_{d2} = 2v_{d1}$.

Golden points

★ *Orders of magnitude: drift* $\sim 10^{-4}\,\mathrm{m\,s}^{-1}$, *thermal* $\sim 10^{5}\,\mathrm{m\,s}^{-1}$, *signal (field)* $\sim 3 \times 10^{8}\,\mathrm{m\,s}^{-1}$.

Mobility and Current Density

1 Mobility

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$

drift speed per unit field · unit $\mathrm{m^2\,V^{-1}\,s^{-1}}$ *· positive by definition*

2 Current density

$$J = \frac{I}{A} = n\,e\,v_d$$

A perpendicular to the flow, current uniform over A · $\vec{J}$: vector along $\vec{E}$ (conventional current) · $\mathrm{A\,m^{-2}}$ *· in general* $I = \int \vec{J}\cdot d\vec{A}$

3 Ohm's law, microscopic form

$$\vec{J} = \sigma\vec{E}, \qquad \sigma = \frac{ne^2\tau}{m} = \frac{1}{\rho}$$

ohmic conductor at fixed temperature (n, τ constant) · σ: conductivity $(\mathrm{S\,m^{-1}})$ *· ρ: resistivity* $(\Omega\,\mathrm{m})$

Example

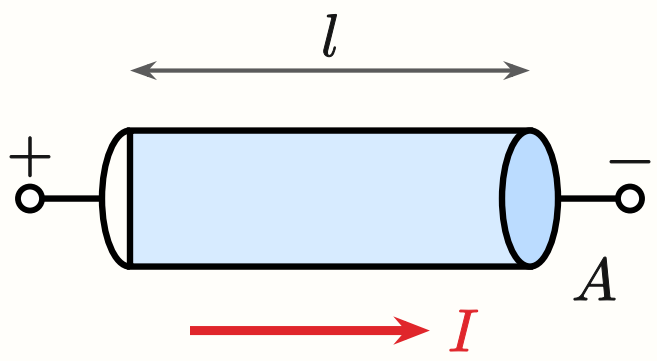
$n = 8.5\times10^{28}\,\mathrm{m^{-3}}$, $\tau = 2.5\times10^{-14}\,\mathrm{s}$: $\mu \approx 4.4\times10^{-3}\,\mathrm{m^2\,V^{-1}\,s^{-1}}$, $\sigma \approx 6.0\times10^{7}\,\mathrm{S\,m^{-1}}$, $\rho \approx 1.7\times10^{-8}\,\Omega\,\mathrm{m}$. *In* $E = 0.025\,\mathrm{V\,m^{-1}}$: $v_d = \mu E \approx 1.1\times10^{-4}\,\mathrm{m\,s^{-1}}$.

Golden points

★ *Uniform wire:* $V = El$ *and* $I = JA$, *so* $J = \sigma E$ *gives* $V = Il/\sigma A$, *i.e.* $R = \rho l/A$.

Ohm's Law and Resistance

1 Ohm's law



$$V = IR$$

$V \propto I$ only while temperature and strain stay the same · $1\,\Omega = 1\,\mathrm{V\,A^{-1}}$ · $[\mathrm{ML^2T^{-3}A^{-2}}]$

2 Resistance from the dimensions

$$R = \frac{\rho\, l}{A}$$

uniform conductor, current along its length l, cross-section A · ρ depends on the material and temperature, NOT on the shape · $\Omega\,\mathrm{m}$, $[\mathrm{ML^3T^{-3}A^{-2}}]$

Example

Copper, $l = 10\,\mathrm{m}$, $A = 1.0\,\mathrm{mm^2}$, $\rho = 1.7 \times 10^{-8}\,\Omega\,\mathrm{m}$: $R = 0.17\,\Omega$, $G = 1/R \approx 5.9\,\mathrm{S}$.

3 Conductance and conductivity

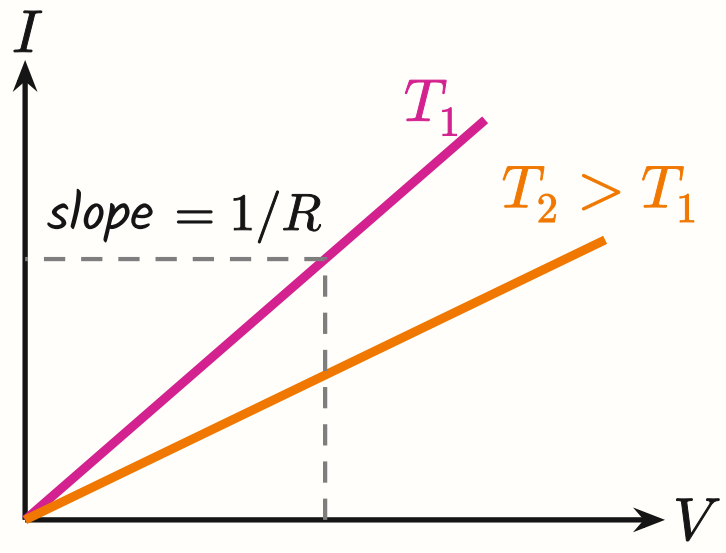
- $G = 1/R$ in siemens ($1\,\mathrm{S} = 1\,\Omega^{-1}$, *mho*); $\sigma = 1/\rho$ in $\mathrm{S\,m^{-1}}$.
- ρ at room temperature: *metals* 10^{-8}–$10^{-6}\,\Omega\,\mathrm{m}$; *insulators* 10^{11}–$10^{19}\,\Omega\,\mathrm{m}$; *semiconductors in between*.

Golden points

★ Block $1 \times 2 \times 4\,\mathrm{cm}$, electrodes on opposite faces: $R_{max}/R_{min} = 16$ ($l = 4\,\mathrm{cm}$ on $2\,\mathrm{cm^2}$ vs $l = 1\,\mathrm{cm}$ on $8\,\mathrm{cm^2}$).

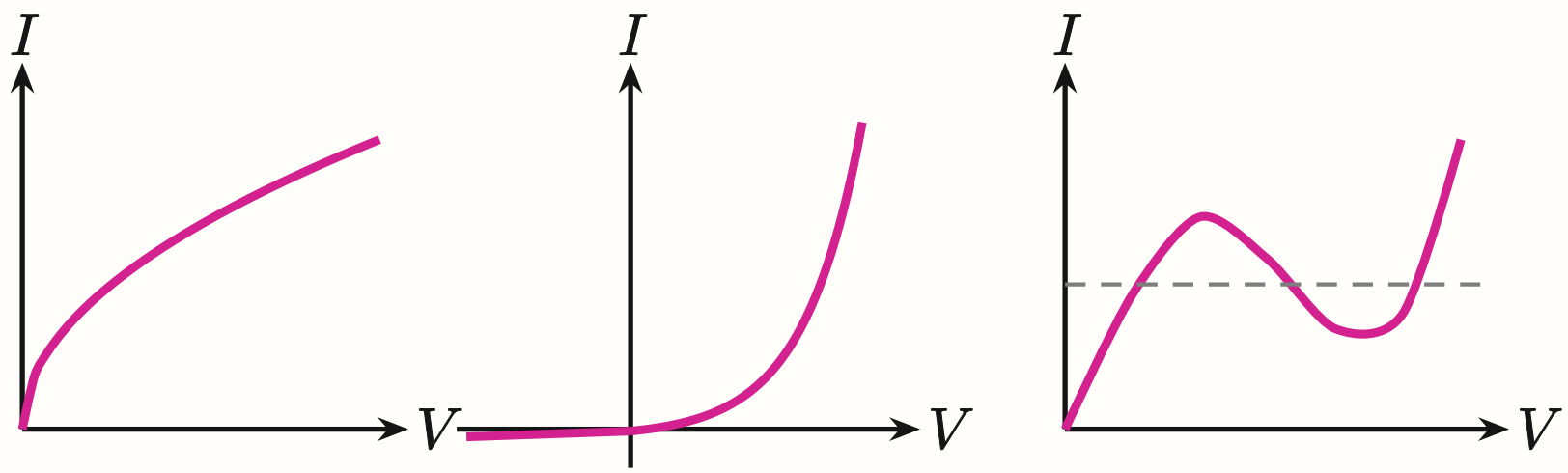
V–I Graphs

1 Ohmic conductor



- Straight line through the origin: R constant.
- Slope of the I–V graph $= 1/R$; slope of a V–I graph $= R$.
- Metal, hotter (T_2): R larger, I–V line less steep.

2 Non-ohmic devices



(a) filament lamp (b) diode (c) GaAs

- (a) Curved: R rises as the filament heats up.
- (b) Depends on the SIGN of V: almost no current in reverse.
- (c) The same I at more than one V (dashed line).

Golden points

★ Any device: $R = V/I$ at that point; dynamic $r = dV/dI$. Equal only for an ohmic conductor. Lamp: $6\,\text{V}$, $0.5\,\text{A}$: $R = 12\,\Omega$; $1\,\text{V}$, $0.2\,\text{A}$: $R = 5\,\Omega$ (cooler filament).

Resistivity and Temperature

1 Linear law

$$\rho_T = \rho_0 \left[1 + \alpha \left(T - T_0\right)\right]$$

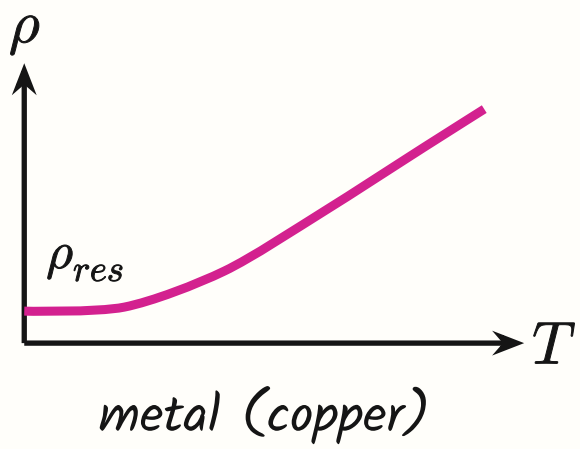
$$R_T = R_0 \left[1 + \alpha \left(T - T_0\right)\right]$$

$\alpha = \frac{1}{\rho_0}\frac{d\rho}{dT}$: temperature coefficient (K^{-1}) · metals, limited range around T_0 · R form: expansion neglected

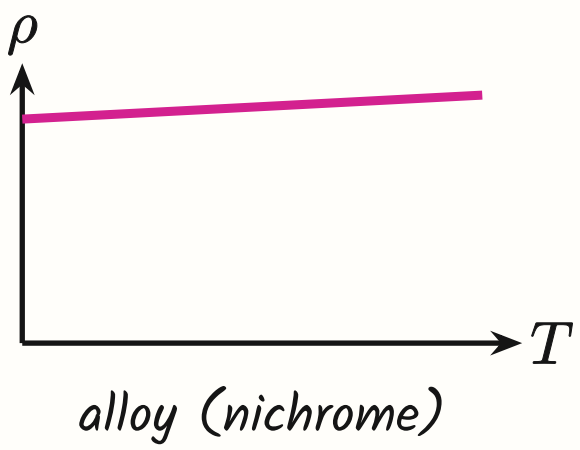
Example

$R_0 = 10\,\Omega$ at $T_0 = 20°\mathrm{C}$, $\alpha = 4.0 \times 10^{-3}\,°\mathrm{C}^{-1}$: $14\,\Omega$ at $120°\mathrm{C}$, $12\,\Omega$ at $70°\mathrm{C}$.

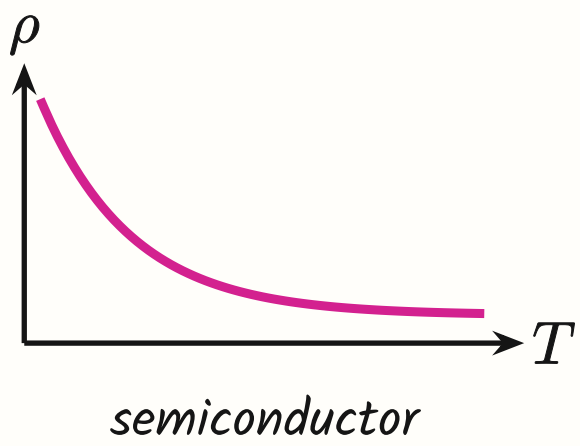
2 ρ versus T



metal (copper)



alloy (nichrome)



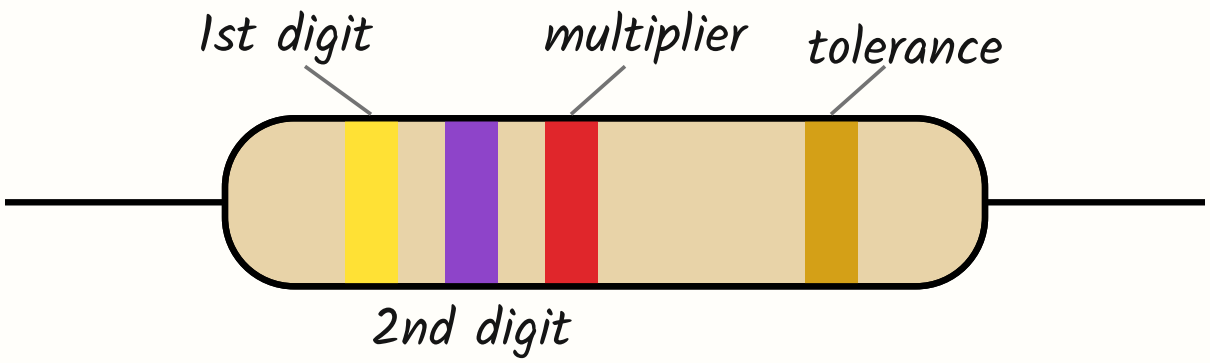
semiconductor

3 Why

- Metal ($\alpha > 0$): n hardly changes; ions vibrate more, τ falls, $\rho = m/ne^2\tau$ rises (levels off at a residual value near $0\,\mathrm{K}$).
- Semiconductor ($\alpha < 0$): n rises steeply with T, outweighing the fall in τ. Thermistor: large negative α.
- Alloys: large ρ, tiny α. Superconductor: $\rho = 0$ below a critical T (mercury $4.2\,\mathrm{K}$).

Colour Code and Materials

1 Colour code of carbon resistors



$$R = (AB) \times 10^{C}\,\Omega \ \pm D$$

A, B: digits of bands 1 and 2 · C: band 3 · D: band 4 (gold $\pm 5\%$, silver $\pm 10\%$, no band $\pm 20\%$)

Black	0	Green	5
Brown	1	Blue	6
Red	2	Violet	7
Orange	3	Grey	8
Yellow	4	White	9
Gold	$\times 10^{-1}$	Silver	$\times 10^{-2}$

Example

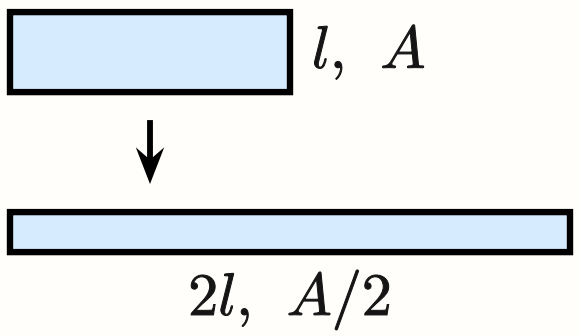
Yellow, violet, red, gold: $47 \times 10^{2}\,\Omega = 4.7\,\mathrm{k}\Omega \pm 5\%$ (4465 to $4935\,\Omega$). Brown, black, orange, silver: $10\,\mathrm{k}\Omega \pm 10\%$.

Golden points

- ★ Digits 0–9: "B B ROY of Great Britain has Very Good Wife".
- ★ Manganin, constantan (tiny α): standard resistors, bridge and potentiometer wires. Nichrome (high ρ and melting point): heaters. Tungsten: filaments. Copper: connecting wires.

Stretching, Cutting, Shapes

1 Stretching at fixed volume



$$R = \frac{\rho\, l^2}{V} \propto l^2 \propto \frac{1}{r^4}$$

volume $V = Al$ fixed · same material and temperature (ρ unchanged) · uniform wire

Example

$10\,\Omega$ wire stretched to twice its length: $40\,\Omega$. Drawn out until its radius halves: $16\times$, $160\,\Omega$. Small stretch: $\Delta R/R = 2\,\Delta l/l$ (1% longer: R up about 2%).

2 Cutting and bending

- *Cut into n equal parts: each R/n; all n joined in parallel: R/n^2. $10\,\Omega$, $n = 5$: $2\,\Omega$ each, $0.4\,\Omega$ together.*
- *Uniform wire of $16\,\Omega$ bent into a ring: the two arcs are in parallel. Diametrically opposite points: $8 \parallel 8 = 4\,\Omega$; a quarter arc: $4 \parallel 12 = 3\,\Omega$.*

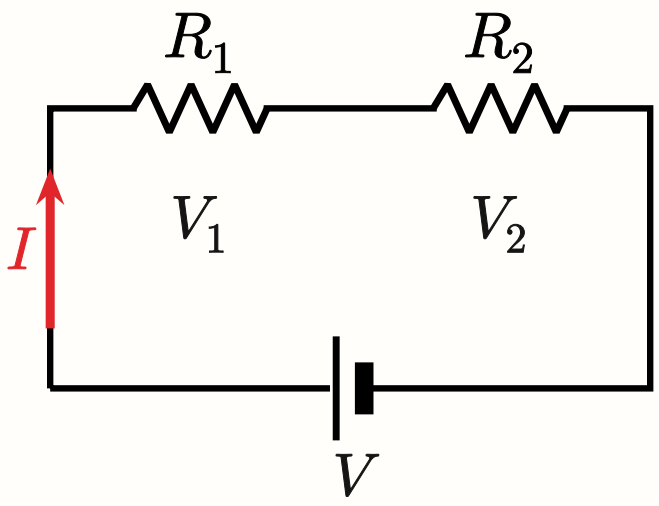
3 Non-uniform cross-section

$$R = \int_0^L \frac{\rho\, dx}{A(x)}, \qquad R_{cone} = \frac{\rho L}{\pi a b}$$

current along the axis, spread evenly over each cross-section (gentle taper) · truncated cone of length L, end radii a and b

Series and Parallel

1 Series



$$R_s = R_1 + R_2 + \dots$$

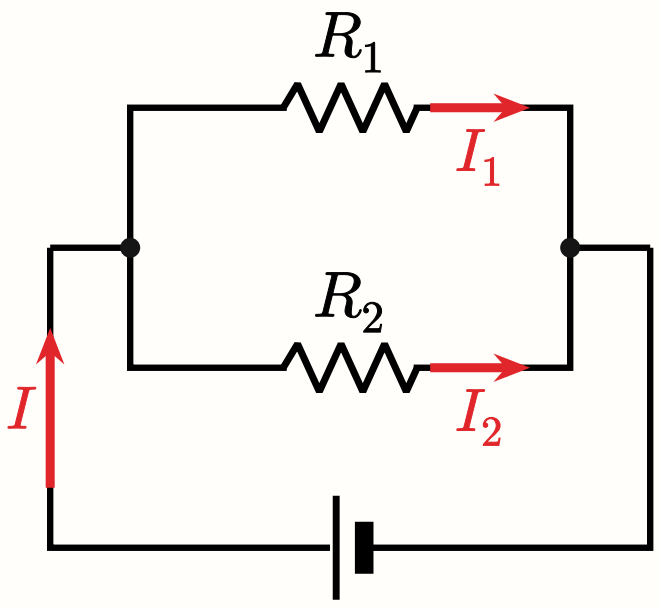
$$V_1 = \frac{R_1}{R_1 + R_2} V$$

the SAME current through each · V divides in proportion to R

Example

$2\,\Omega + 4\,\Omega$ *on an ideal* $12\,\mathrm{V}$ *cell:* $I = 2\,\mathrm{A}$, $V_1 = 4\,\mathrm{V}$, $V_2 = 8\,\mathrm{V}$.

2 Parallel



$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$I_1 = \frac{R_2}{R_1 + R_2} I$$

the SAME V across each · I divides inversely as R

Example

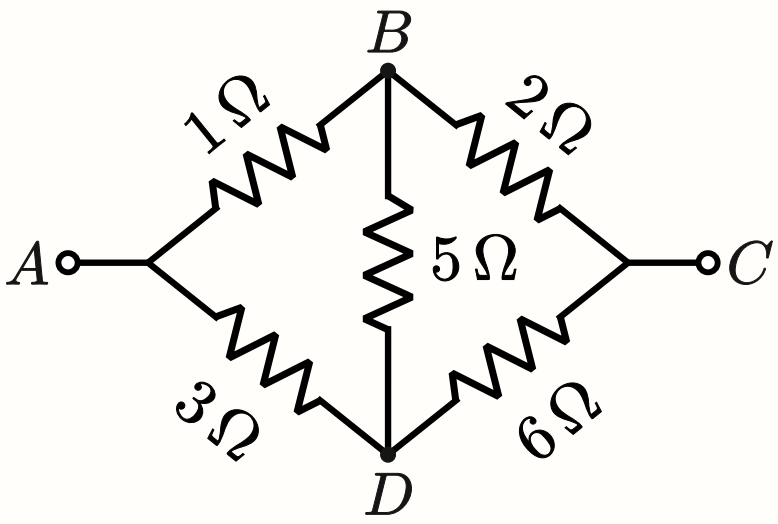
$6\,\mathrm{A}$ *into* $2\,\Omega \parallel 4\,\Omega$: $R_p = 4/3\,\Omega$, $I_1 = 4\,\mathrm{A}$ *(through* $2\,\Omega$*)*, $I_2 = 2\,\mathrm{A}$.

Golden points

★ $R_s >$ *largest,* $R_p <$ *smallest (*2, 3, $6\,\Omega$: $11\,\Omega$ *and* $1\,\Omega$*).* n *identical:* nR *and* R/n*, ratio* n^2.

Symmetry and Balanced Bridges

1 Balanced Wheatstone bridge



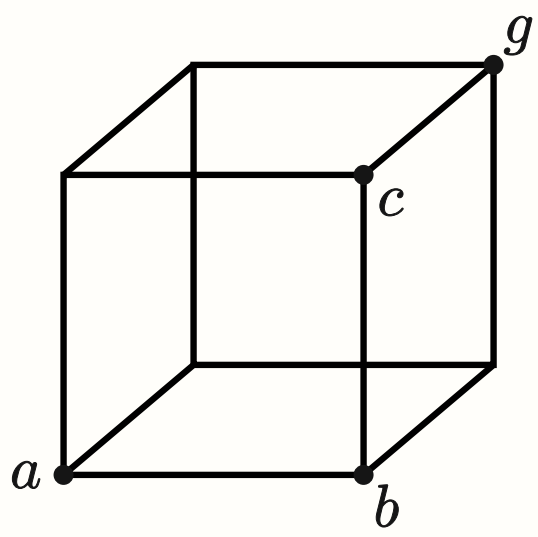
$$\frac{P}{Q} = \frac{R}{S} \Rightarrow I_{BD} = 0$$

P, Q: upper arms; R, S: lower arms · any resistor (or meter) in BD carries no current

Example

$\frac{1}{2} = \frac{3}{6}$; balanced;
$R_{AC} = (1+2) \parallel (3+6) = 2.25\,\Omega$,
whatever is in BD.

2 Cube of 12 equal resistors R



- Edge ab: $\frac{7}{12}R$.
- Face diagonal ac: $\frac{3}{4}R$.
- Body diagonal ag: $\frac{5}{6}R$.
- $R = 12\,\Omega$: 7, 9, 10 Ω.

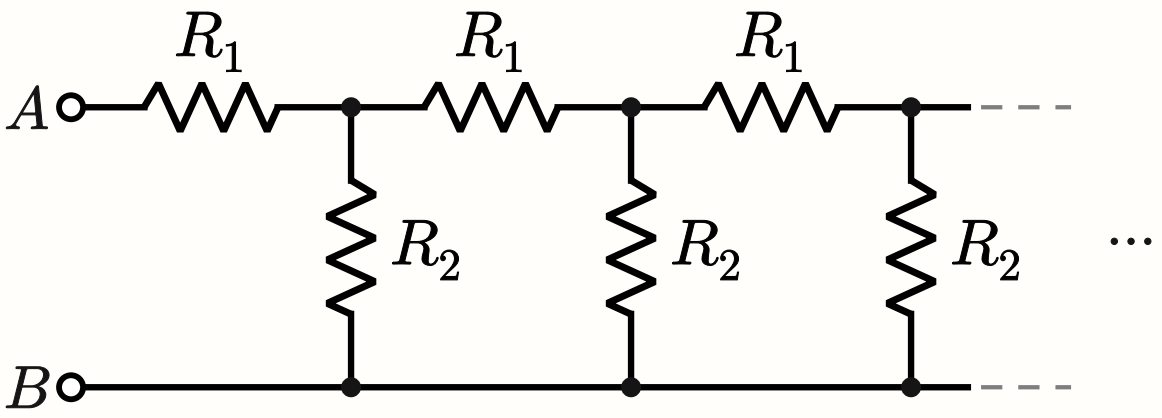
Golden points

★ A resistor between two EQUAL-potential points carries no current: remove it. Identical resistors: points symmetric about the input-output line are at equal potential.

★ Body diagonal: 3 nearest corners at one potential, 3 farthest at another: $\frac{R}{3} + \frac{R}{6} + \frac{R}{3} = \frac{5}{6}R$.

Ladders and Tricky Networks

1 Infinite ladder



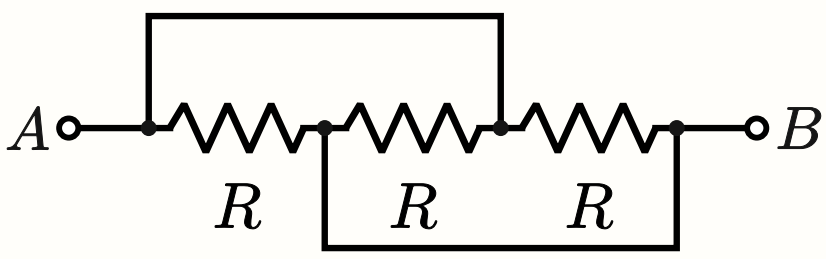
$$R_x = R_1 + \frac{R_2 R_x}{R_2 + R_x} \;\Rightarrow\; R_x = \frac{R_1 + \sqrt{R_1^2 + 4R_1R_2}}{2}$$

infinitely many identical sections: adding one leaves $R_{AB} = R_x$ unchanged

Example

$R_1 = 1\,\Omega$, $R_2 = 2\,\Omega$: $R_x = 2\,\Omega$. $R_1 = R_2 = R$:
$R_x = \frac{1+\sqrt{5}}{2} R \approx 1.618\,R$.

2 Shorting wires: label the nodes



- A plain wire joins points at ONE potential (zero resistance).
- Here every resistor goes from A to B: $R_{AB} = R/3$.

Golden points

★ Infinite square grid of equal resistors R: between ADJACENT nodes $R/2$ (superpose I fed in and I taken out: $I/4 + I/4$ in the link).

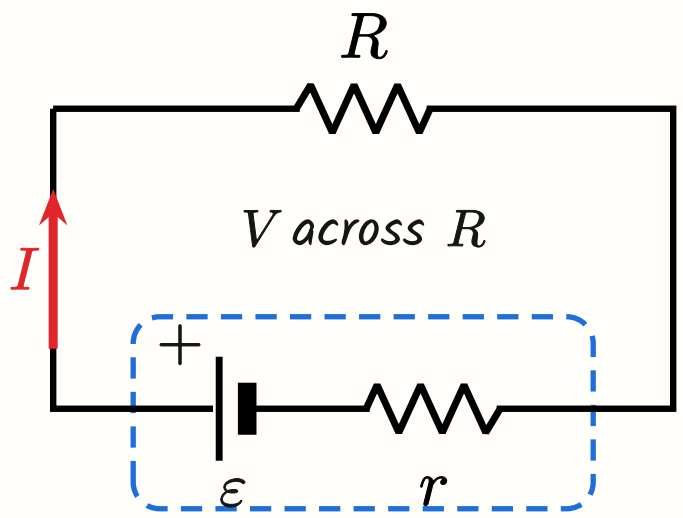
EMF and Internal Resistance

1 Electromotive force

$$\varepsilon = \frac{W}{q}$$

W: work by the cell's non-electrostatic (chemical) force on q taken from $-$ to $+$ inside · volt · $=$ terminal p.d. on OPEN circuit

2 Cell supplying current (discharging)



$$I = \frac{\varepsilon}{R + r}$$

$$V = \varepsilon - Ir = IR$$

current LEAVES the $+$ terminal (cell gives energy) · r: internal resistance (electrolyte) · V: terminal p.d.

Example

$\varepsilon = 12\,\text{V}$, $r = 0.5\,\Omega$, $R = 5.5\,\Omega$: $I = 2\,\text{A}$, $V = 11\,\text{V}$ ("lost volts" $Ir = 1\,\text{V}$).

3 Energy per second

$$\varepsilon I = I^2 R + I^2 r$$

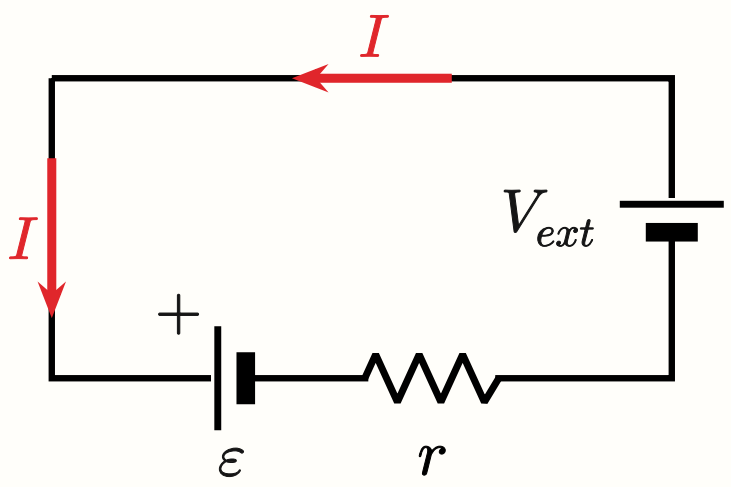
same example: $24\,\text{W}$ from the cell $= 22\,\text{W}$ in R + $2\,\text{W}$ heating the cell

Golden points

★ Open circuit: $V = \varepsilon$. Short circuit ($R = 0$): $I = \varepsilon / r = 24\,\text{A}$ here, $V = 0$. Ideal cell ($r = 0$): $V = \varepsilon$ always.

Charging and the Cell Graph

1 Cell being charged



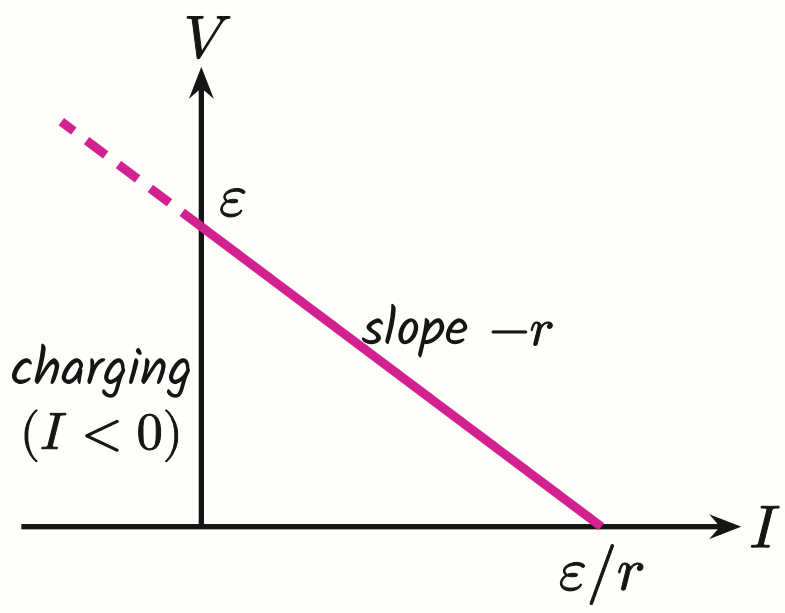
$$V = \varepsilon + Ir$$

current ENTERS the + terminal (an external source drives it) · $V > \varepsilon$

Example

$\varepsilon = 12\,\text{V}$, $r = 0.5\,\Omega$, *charged at* $2\,\text{A}$: $V = 13\,\text{V}$; $26\,\text{W}$ *in* $= 24\,\text{W}$ *stored* $+ 2\,\text{W}$ *heat.*

2 Terminal voltage versus current



$$V = \varepsilon - Ir$$

ε, r constant · intercepts ε and $I_{max} = \varepsilon/r$ (short circuit)

Example

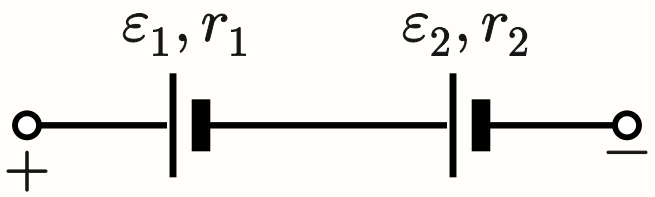
$V = 1.4\,\text{V}$ *at* $0.2\,\text{A}$ *and* $V = 1.2\,\text{V}$ *at* $0.6\,\text{A}$: $r = 0.5\,\Omega$, $\varepsilon = 1.5\,\text{V}$.

Golden points

★ $V < \varepsilon$: *the cell is supplying current.* $V > \varepsilon$: *it is being charged.* $V = \varepsilon$: *no current (or an ideal cell).*

Cells in Series and Parallel

1 Series



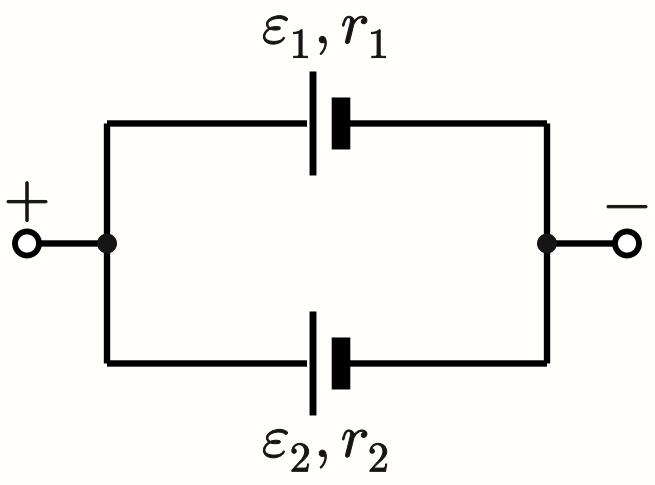
$$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2, \quad r_{eq} = r_1 + r_2$$

+ of one cell to − of the next · a REVERSED cell enters with $-\varepsilon$ (its r still adds)

Example

4 cells ($1.5\,\mathrm{V}$, $0.25\,\Omega$ each) in series, $R = 5\,\Omega$: $6\,\mathrm{V}$, $1\,\Omega$, $I = 1\,\mathrm{A}$.
One cell reversed: $3\,\mathrm{V}$, $1\,\Omega$, $I = 0.5\,\mathrm{A}$.

2 Parallel



$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

+ terminals joined together (one reversed: change the sign of its ε)

Example

$6\,\mathrm{V}$, $1\,\Omega$ in parallel with $3\,\mathrm{V}$, $2\,\Omega$: $\varepsilon_{eq} = 5\,\mathrm{V}$, $r_{eq} = 2/3\,\Omega$.

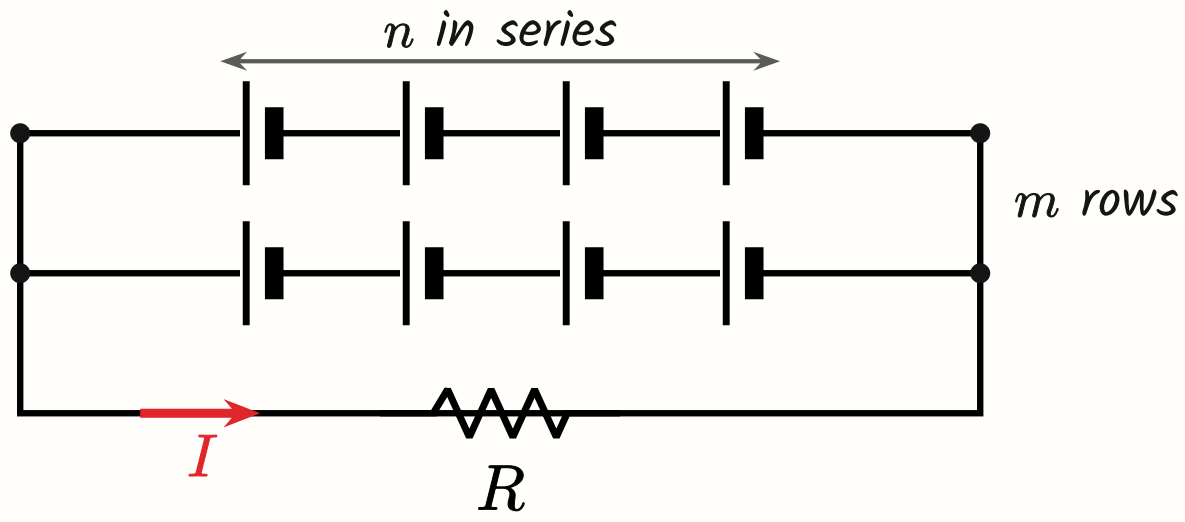
Golden points

★ n identical cells in parallel: $\varepsilon_{eq} = \varepsilon$, $r_{eq} = r/n$, so $I = \dfrac{\varepsilon}{R + r/n}$.

★ Unequal cells in parallel drive a current round their own loop even with no load: avoid it in practice.

Mixed Grouping of Cells

1 m rows of n cells



$$I = \frac{n\varepsilon}{R + nr/m}$$

$N = mn$ identical cells (ε, r) · n in series per row, m rows in parallel

2 Maximum current

$$\frac{nr}{m} = R \;\Rightarrow\; I_{max} = \frac{\varepsilon}{2}\sqrt{\frac{N}{rR}}$$

N fixed · external R = battery's internal resistance

Example

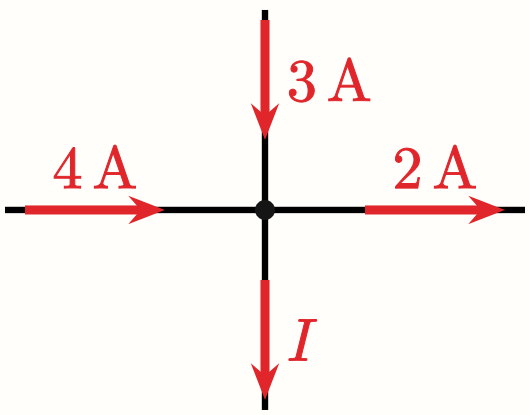
$N = 8$, $\varepsilon = 2\,\text{V}$, $r = 1\,\Omega$, $R = 2\,\Omega$ $(n \times m)$: 8×1: $1.6\,\text{A}$; 4×2: $2.0\,\text{A}$ (best, $nr/m = 2\,\Omega = R$); 2×4: $1.6\,\text{A}$; 1×8: $0.94\,\text{A}$.

Golden points

★ $R \geq Nr$: all cells in series. $R \leq r/N$: all in parallel. Otherwise $nr/m = R$; there the power in R is also maximum.

Kirchhoff's Laws

1 Junction rule



$$\sum I_{in} = \sum I_{out}$$

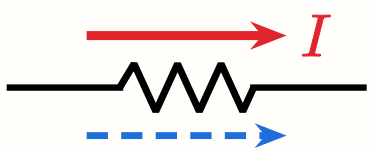
conservation of charge · steady state (no charge piles up) · here $4 + 3 = 2 + I$*:* $I = 5\,\mathrm{A}$

2 Loop rule

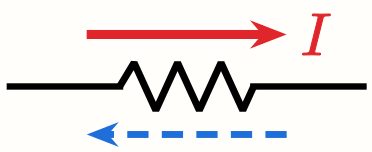
$$\sum \Delta V = 0 \quad \textit{round any closed loop}$$

conservation of energy: the electrostatic field is conservative · steady currents

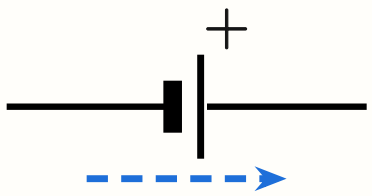
3 Signs (blue dashes: walking direction)



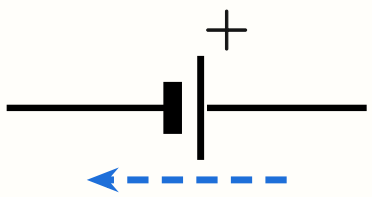
resistor, walking ALONG I $\quad -IR$



resistor, AGAINST I $\quad +IR$



cell, from $-$ *to* $+$ $\quad +\varepsilon$



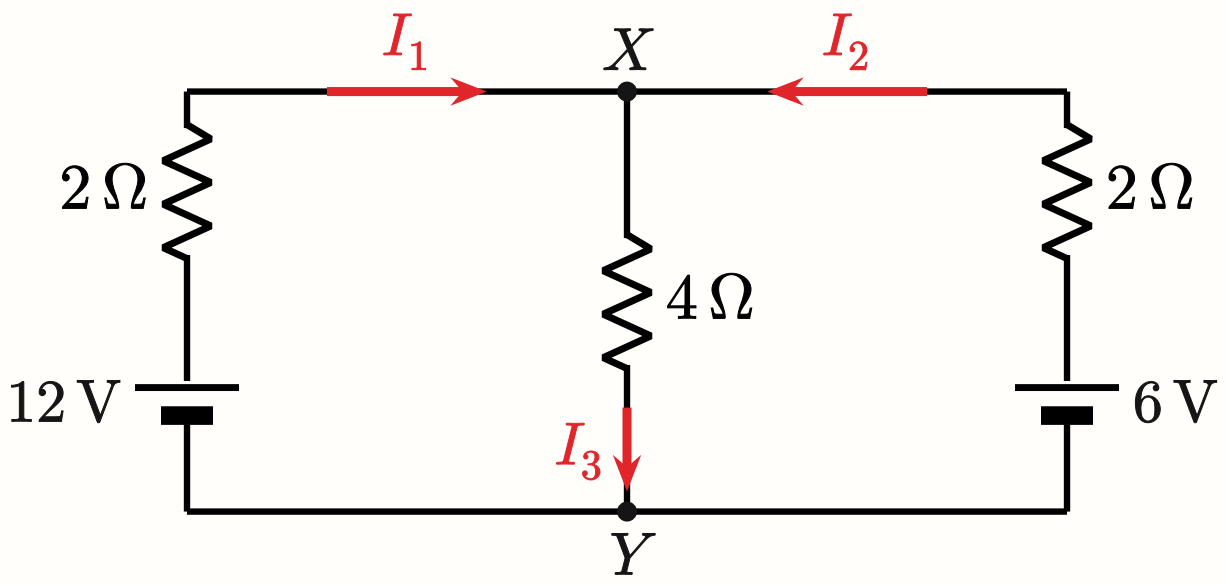
cell, from $+$ *to* $-$ $\quad -\varepsilon$

Golden points

★ *Guess any directions for the unknown currents; a NEGATIVE answer means the current flows against the guessed arrow.*

Kirchhoff: Worked Example

1 Two cells, three branches



$$12 - 2I_1 - 4I_3 = 0, \qquad 6 - 2I_2 - 4I_3 = 0$$

$$I_1 + I_2 = I_3$$

loops walked up through each cell (− to +) · junction X

Example

$I_3 = 1.8\,\text{A}$, $I_1 = 2.4\,\text{A}$, $I_2 = -0.6\,\text{A}$: $0.6\,\text{A}$ flows DOWN through the $6\,\text{V}$ cell, into its + terminal: it is being charged.

2 Potential difference between two points

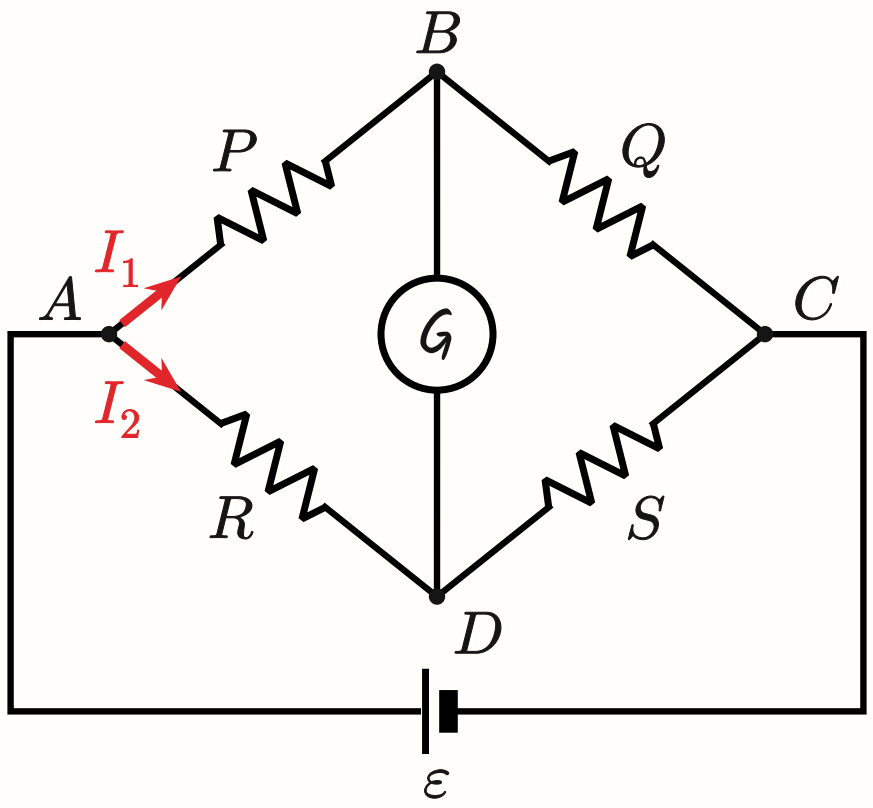
- $V_X - V_Y$: add the changes walking from Y to X along ANY path; every path gives the same answer.
- Middle: $4 \times 1.8 = 7.2\,\text{V}$. Left: $12 - 2 \times 2.4 = 7.2\,\text{V}$. Right: $6 + 2 \times 0.6 = 7.2\,\text{V}$ ($V = \varepsilon + IR_{branch}$, cell charging).

Golden points

★ Nodal method: take $V_Y = 0$ and add the currents leaving X: $\frac{V-12}{2} + \frac{V}{4} + \frac{V-6}{2} = 0$ gives $V = 7.2\,\text{V}$ in one line.

Wheatstone Bridge

1 Balance condition



$$\frac{P}{Q} = \frac{R}{S}$$

galvanometer shows NO deflection (null point): $V_B = V_D$ · cell across AC, galvanometer across BD

- $I_g = 0$: P, Q carry I_1; R, S carry I_2. $V_B = V_D$: $I_1 P = I_2 R$, $I_1 Q = I_2 S$; divide.

Example

$P = 10\,\Omega$, $Q = 20\,\Omega$, $R = 15\,\Omega$, *null deflection:* $S = 30\,\Omega$.

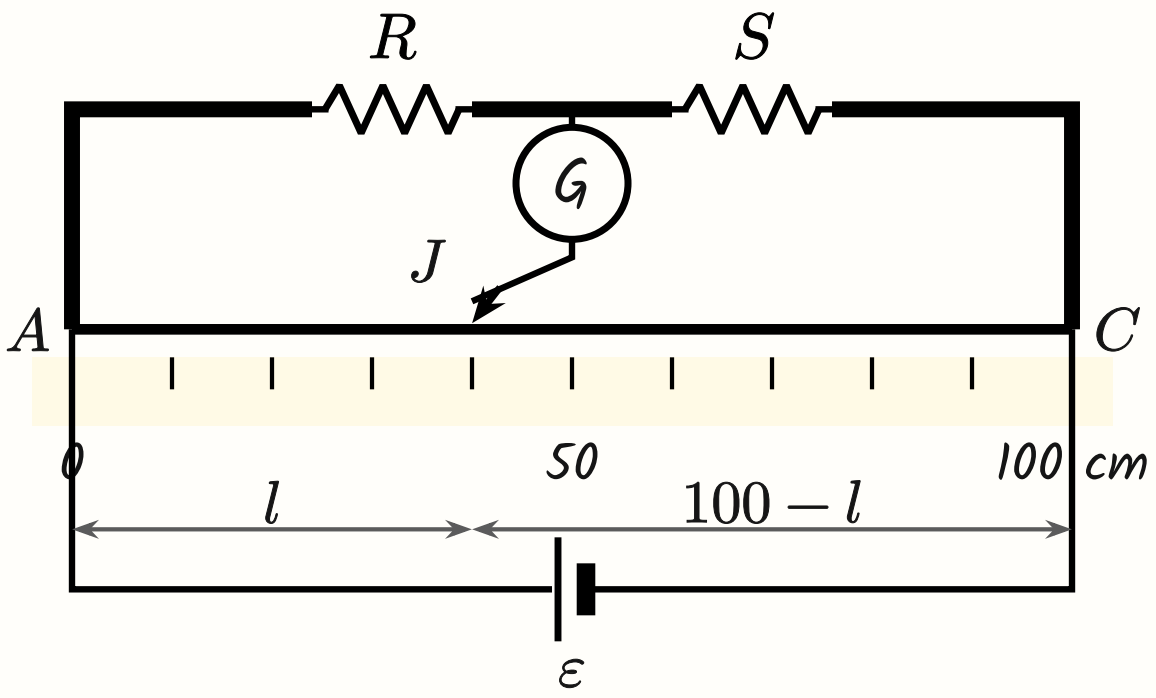
Golden points

★ *Balance is independent of the cell's ε, r and of G; cell and galvanometer can be interchanged.*

★ *Most sensitive when P, Q, R, S are of the same order. Unbalanced bridge: use Kirchhoff's laws.*

Metre Bridge

1 A Wheatstone bridge with a wire



$$\frac{R}{S} = \frac{l}{100 - l}$$

l (cm) from A, the end next to R · uniform wire · strips of negligible R

Example

$S = 10\,\Omega$, $l = 40\,\text{cm}$: $R = 10 \times 40/60 \approx 6.67\,\Omega$; *swapped:* $60\,\text{cm}$.

2 End correction

$$\frac{R}{S} = \frac{l + \alpha}{100 - l + \beta}$$

α, β: extra wire lengths equivalent to the joint resistances at A and C

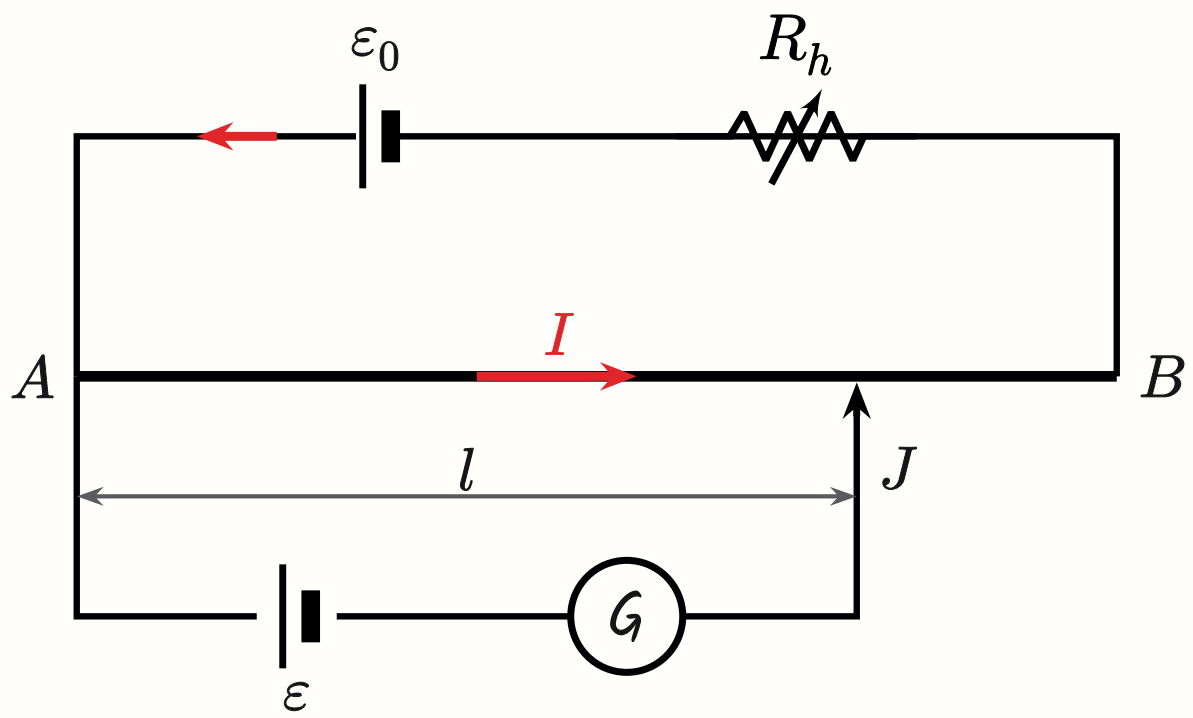
Golden points

★ *Best accuracy: balance near the middle (40–60 cm); swap R, S and average.*

Potentiometer: Principle

Older-syllabus topic: check your exam's current syllabus

1 Uniform wire, steady current



$$V_{AJ} = k\,l, \qquad k = \frac{V_{AB}}{L}$$

k: potential gradient ($\mathrm{V\,m^{-1}}$) · wire of uniform cross-section and material, length L · steady current in the primary

$$\varepsilon = k\,l_0 \quad \text{(balance)}$$

null point: no current through ε, so the EMF itself is measured

Example

$\varepsilon_0 = 2\,\mathrm{V}$ (r negligible), $R_h = 2\,\Omega$, wire $L = 4\,\mathrm{m}$ of $8\,\Omega$: $I = 0.2\,\mathrm{A}$, $V_{AB} = 1.6\,\mathrm{V}$, $k = 0.4\,\mathrm{V\,m^{-1}}$; a $1.2\,\mathrm{V}$ cell balances at $3.0\,\mathrm{m}$.

Golden points

★ No balance point unless $V_{AB} \geq \varepsilon$ AND the $+$ terminals of both cells go to the SAME end A.

Potentiometer: Uses

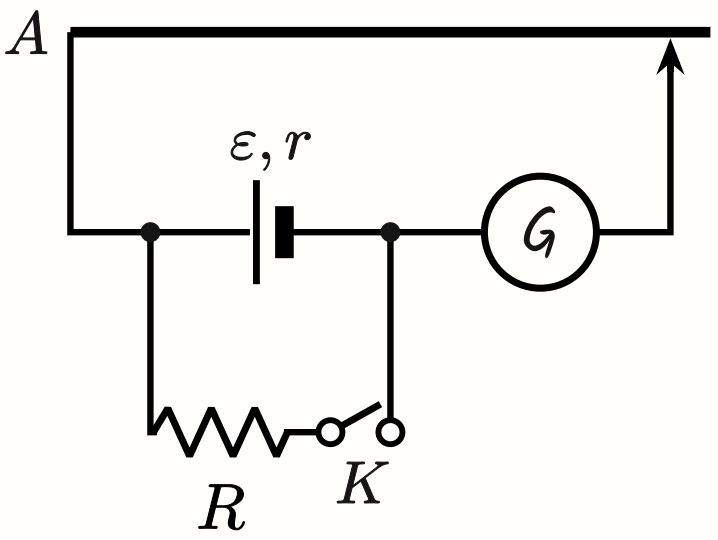
Older-syllabus topic: check your exam's current syllabus

1 Comparing two EMFs

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$$

same primary current (same k) · $1.2\,\mathrm{V}$ at $3.0\,\mathrm{m}$, ε_2 at $2.5\,\mathrm{m}$: $\varepsilon_2 = 1.0\,\mathrm{V}$

2 Internal resistance of a cell



$$r = R\left(\frac{l_1}{l_2} - 1\right)$$

l_1: balance with K open (ε) · l_2: K closed (terminal p.d. V across R)

Example

$l_1 = 60\,\mathrm{cm}$, $l_2 = 50\,\mathrm{cm}$, $R = 10\,\Omega$: $r = 10(1.2 - 1) = 2\,\Omega$.

3 Sensitivity

- *Smaller k (longer wire, or more series resistance in the primary): finer reading. $k = 0.4\,\mathrm{V\,m^{-1}}$ read to $1\,\mathrm{mm}$: $0.4\,\mathrm{mV}$.*

Golden points

★ *At balance it draws NO current from the cell (an ideal voltmeter); a real voltmeter draws current and reads $V < \varepsilon$.*

Electrical Energy and Power

1 Energy and power

$$W = VIt, \qquad P = VI$$

ANY device (resistor, motor, charging cell) · $1\,\mathrm{W} = 1\,\mathrm{J\,s^{-1}}$

$$P = I^2R = \frac{V^2}{R}$$

only a pure resistance (all energy becomes heat); not a motor or a charging cell

2 Heating effect (Joule's law)

$$H = I^2Rt$$

steady current I · R constant · H in joule

Example

$1000\,\mathrm{W}$ *kettle at its rated voltage,* $1\,\mathrm{kg}$ *water from* $20^\circ\mathrm{C}$ *to* $100^\circ\mathrm{C}$ *(*$c = 4200\,\mathrm{J\,kg^{-1}\,K^{-1}}$*, no heat lost):* $H = 336\,\mathrm{kJ}$, $t = 336\,\mathrm{s}$.

3 Commercial unit

- $1\,\mathrm{kWh}$ *(one "unit")* $= 3.6 \times 10^6\,\mathrm{J}$. *A* $1.5\,\mathrm{kW}$ *heater used* $2\,\mathrm{h}$ *a day for* 30 *days:* $90\,\mathrm{kWh}$.

Golden points

★ *Transmission line of resistance* R_L*: loss* $= (P/V)^2R_L$. $100\,\mathrm{kW}$ *sent at* $1000\,\mathrm{V}$ *through* $0.5\,\Omega$*:* $5\,\mathrm{kW}$ *lost; at* $10\,\mathrm{kV}$*: only* $50\,\mathrm{W}$.

★ *Fuse: wire of low melting point in SERIES with the line; it melts when the current exceeds its rating.*

Rated Power and Bulbs

1 Rated values

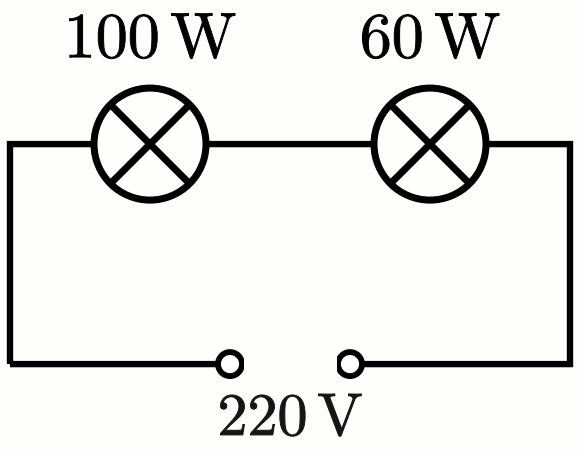
$$R = \frac{V_r^2}{P_r}, \qquad P = P_r \left(\frac{V}{V_r}\right)^2$$

P_r at the rated voltage V_r · R assumed constant (filament heating ignored)

Example

$100\,\mathrm{W}$, $220\,\mathrm{V}$: $R = 484\,\Omega$; $60\,\mathrm{W}$, $220\,\mathrm{V}$: $R \approx 807\,\Omega$. *The* $100\,\mathrm{W}$ *bulb on* $110\,\mathrm{V}$: $25\,\mathrm{W}$.

2 Two bulbs in series



- *Same* I: $P = I^2R \propto R$. *The LOWER-rated* $60\,\mathrm{W}$ *bulb (larger* R*) glows brighter.*
- *On* $220\,\mathrm{V}$: $P_{100} \approx 14.1\,\mathrm{W}$, $P_{60} \approx 23.4\,\mathrm{W}$, *total* $37.5\,\mathrm{W}$.

$$\frac{1}{P_s} = \frac{1}{P_1} + \frac{1}{P_2}$$

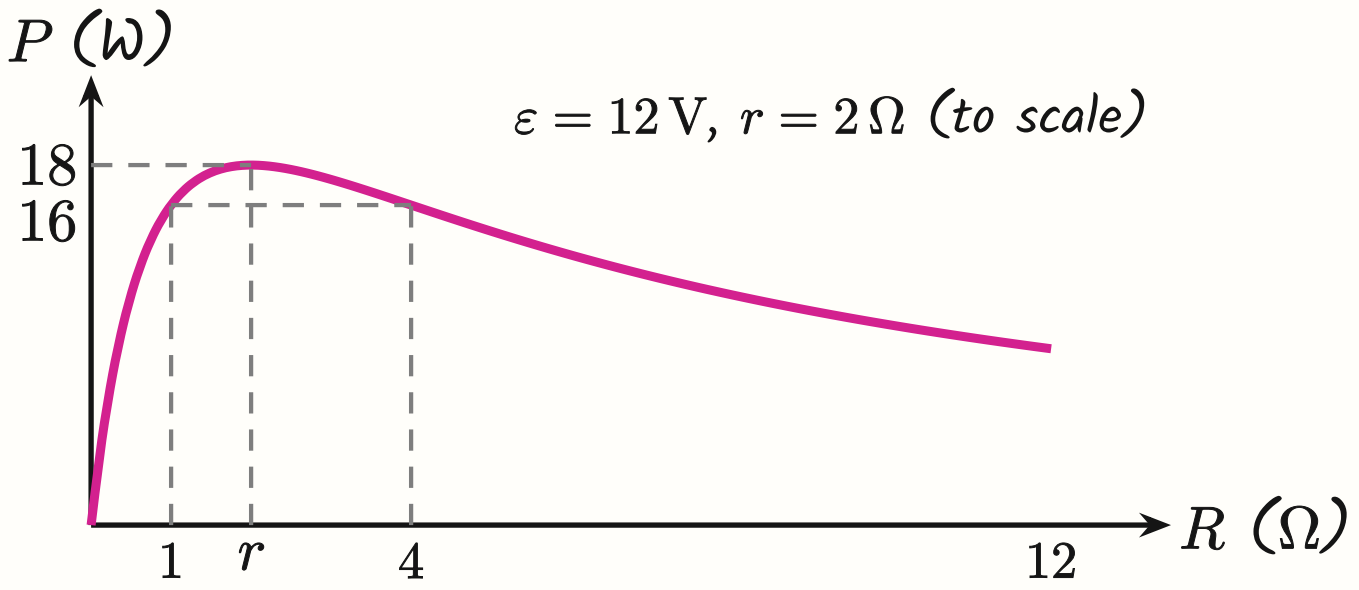
same rated voltage, supply = rated voltage, R constant

3 Two bulbs in parallel

- *Same* V: $P \propto 1/R$. *On the rated voltage each gives its rating:* $160\,\mathrm{W}$ *in all;* $100\,\mathrm{W}$ *brighter. Brightness follows the power actually dissipated.*

Maximum Power Transfer

1 Power in the load R



$$P = \frac{\varepsilon^2 R}{(R+r)^2}, \qquad P_{max} = \frac{\varepsilon^2}{4r} \text{ at } R = r$$

source ε, r fixed, load R varied

Example

$\varepsilon = 12\,\text{V}$, $r = 2\,\Omega$: $P_{max} = 18\,\text{W}$ at $R = 2\,\Omega$; $R = 1\,\Omega$ and $R = 4\,\Omega$ both give $16\,\text{W}$.

2 Efficiency

$$\eta = \frac{I^2 R}{I^2(R+r)} = \frac{R}{R+r}$$

at maximum power ($R = r$): $\eta = 50\%$ · $\eta \to 100\%$ only as $R \to \infty$, where $P \to 0$

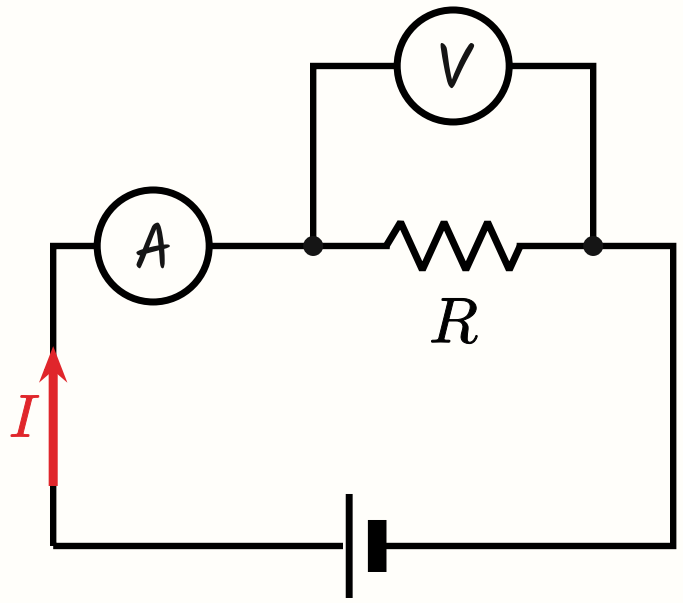
Golden points

★ Two loads giving the same power: $R_1 R_2 = r^2$ ($1 \times 4 = 2^2$).

★ R fixed, r variable: power in R is largest at $r = 0$, NOT $r = R$.

Ammeter and Voltmeter

1 Connections



- Ammeter in SERIES; ideal: zero resistance. Put in parallel by mistake: large current, may burn out.
- Voltmeter in PARALLEL; ideal: infinite resistance. Put in series: $I \approx 0$, reads nearly ε.

2 Loading errors

Example

Two $100\,\Omega$ in series on an ideal $10\,\mathrm{V}$ cell: a $100\,\Omega$ voltmeter across one reads $3.33\,\mathrm{V}$, not $5\,\mathrm{V}$. A $1\,\Omega$ ammeter in series with $4\,\Omega$ on an ideal $10\,\mathrm{V}$ cell reads $2\,\mathrm{A}$, not $2.5\,\mathrm{A}$.

3 Galvanometer converted

$$S = \frac{I_g G}{I - I_g} \quad (\textit{shunt}), \qquad R_s = \frac{V}{I_g} - G$$

G: coil resistance, I_g: full-scale current · ammeter (range I): S in PARALLEL · voltmeter (range V): R_s in SERIES

Example

$G = 50\,\Omega$, $I_g = 2\,\mathrm{mA}$: 0–$1\,\mathrm{A}$ ammeter, $S \approx 0.1\,\Omega$; 0–$10\,\mathrm{V}$ voltmeter, $R_s = 4950\,\Omega$.

Common Traps

1 Exam traps

Avoid these

- ✘ Current is a vector (it adds algebraically at junctions: a scalar).
- ✘ Electrons move at the signal speed (drift is $\sim 10^{-4}\,\mathrm{m\,s^{-1}}$; the field travels near c).
- ✘ ρ depends on length or area (only on material and temperature).
- ✘ Stretched wire: $R \propto l$ (at fixed volume $R \propto l^2$).
- ✘ ρ of a semiconductor rises with T (it falls; $\alpha < 0$).
- ✘ $V = \varepsilon$ always (only on open circuit or for an ideal cell; charging gives $V > \varepsilon$).
- ✘ $P = I^2R$ for a motor (use $P = VI$; I^2R is only its heating).
- ✘ The higher-wattage bulb is always brighter (for the same rated voltage: in series the lower-wattage one wins).
- ✘ Maximum power means maximum efficiency (at $R = r$ it is only 50%).

2 Solving recipe

Steps

- Mark nodes; merge points joined by plain wires; spot symmetry or a balanced bridge.
- Series / parallel where possible; otherwise Kirchhoff or nodal equations.
- Check signs: a negative current flows against the arrow.

Summary

same I or same V? • balanced? • $V = \varepsilon \mp Ir$